

Mohit Kumar Singh

GenAI Backend Engineer · Multi-Agent Systems · LLM Applications · GCP
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PROFESSIONAL SUMMARY

Production-focused GenAI backend engineer with hands-on experience architecting multi-agent LLM systems, RAG pipelines, and secure cloud-native APIs. At Egen, built an AI analytics platform that replaced Looker Studio dashboards with natural-language querying, shipping supervisor agent routing, domain-specific SQL agents, and a secure BigQuery execution layer on GCP. Holds three Google Cloud certifications. Driven by one goal: AI that actually works in production.

EXPERIENCE

Egen · Associate Software Developer — GenAI Backend | Hyderabad, IN · Mar 2025 – Present

- Architected a natural-language workforce analytics platform using Google ADK + Gemini, replacing manual Looker Studio dashboard navigation for C-suite stakeholders.
- Built a supervisor agent that classifies user intent and routes queries to four specialised domain agents like bench, staffing, utilisation, and skills, each with independent BigQuery SQL generation capabilities.
- Engineered a secure SQL execution layer on top of BigQuery: read-only enforcement, DDL/DML blocking, pre-execution safety validation, and structured error handling.
- Developed FastAPI backend services for real-time agent-to-database communication; implemented a Dialogflow CX + Google Chat bridge with session-persistent conversational state.
- Deployed and maintained cloud-native services on GCP (Cloud Run); participated in CI/CD pipelines, Agile sprints, and live production debugging across multi-agent workflows.

PROJECTS

AI-Powered Analytics Multi-Agent System · Google ADK · Gemini · FastAPI · BigQuery · GCP

- Designed the full agent hierarchy: supervisor for intent classification → specialised agents for bench/staffing/utilisation/skills analytics → secure BigQuery SQL execution in real time.
- Built an LLM reasoning layer that interprets SQL results and returns conversational insights, not raw data - making analytics accessible to non-technical leadership.
- Implemented query guardrails (read-only, DDL/DML prevention, safety checks) to ensure zero risk of accidental data mutation in production.

LLM-Powered Document Intelligence Platform (RAG) · Python · Vespa · GCP

- Designed end-to-end RAG pipeline: document parsing → chunking → embedding preparation → vector indexing on Vespa for sub-second semantic search at scale.
- Integrated retrieval layer with FastAPI backend to deliver LLM-grounded answers over enterprise document corpora; validated reliability and observability on GCP.

AI Video Player — Real-Time Subtitle Generation · Python · PyQt6 · OpenAI Whisper · FFmpeg · CUDA

- Built desktop application with multithreaded pipeline: FFmpeg audio extraction → Whisper transcription → real-time subtitle sync, all running concurrently without blocking UI.
- Implemented CUDA GPU-accelerated inference with automatic CPU fallback; profiled and resolved synchronisation bottlenecks, improving subtitle alignment accuracy measurably.

TECHNICAL SKILLS

AI / GenAI: Multi-Agent Systems, RAG Pipelines, LLM APIs, Google ADK, Gemini, OpenAI Whisper, Dialogflow CX, Prompt Engineering, CUDA

Backend & APIs: FastAPI, Flask, Django, RESTful APIs, Microservices, Multithreading, Concurrency, Session Management

Cloud & DevOps: GCP (Cloud Run, BigQuery, Firestore, Pub/Sub), Microsoft Azure, Docker, CI/CD, Linux

Languages: Python (primary) · C++ · JavaScript

Databases: BigQuery, PostgreSQL, Firestore, Vespa (vector search), SQL

CS Fundamentals: Data Structures & Algorithms, OOP, OS, Distributed Systems, Debugging, Performance Profiling

EDUCATION

G.L. Bajaj Institute of Technology and Management | *Greater Noida · 2021 – 2025*

B.Tech — Information Technology First Division

CERTIFICATIONS

- ▶ Google Cloud Certified — Associate Cloud Engineer (ACE) [↗ Verify](#)
- ▶ Google Cloud Certified — Cloud Digital Leader [↗ Verify](#)
- ▶ Google Cloud Certified — Associate Data Practitioner [↗ Verify](#)